# K803Tilt survey development Manual V1.0

## 1. software development instructions

### 1.1 Data communication protocol

#### 1.1.1 Binary message definition

**basic format：**

Header: 3 synchronous bytes plus 25 bytes of header information;

Data: variables;

CRC 32: 32-bit CRC check of all data including header information.

Masthead:

| Field | Filed name | Filed type | description | byte | offset |
|---|---|---|---|---|---|
| 1 | Sync | char | 0xAA | 1 | 0 |
| 2 | Sync | char | 0x44 | 1 | 1 |
| 3 | Sync | char | 0x12 | 1 | 2 |
| 4 | Header length | unchar | Head length | 1 | 3 |
| 5 | Message ID | ushort | Information index number | 2 | 4 |
| 6 | Reserved | | reserved | 1 | 6 |
| 7 | Reserved | | reserved | 1 | 7 |
| 8 | Message length | ushort | Information length , excluding Header and CRC verification | 2 | 8 |
| 9 | Reserved | | reserved | 2 | 10 |
| 10 | Reserved | | reserved | 1 | 12 |
| 11 | Reserved | | reserved | 1 | 13 |
| 12 | Week | ushort | GPS week | 2 | 14 |
| 13 | ms | uint | GPS milliseconds within the week | 4 | 16 |
| 14 | Reserved | | reserved | 4 | 20 |

| | | | | | |
|---|---|---|---|---|---|
| 15 | Reserved | ushort | reserved | 2 | 24 |
| 16 | Receiver S/W Version | ushort | The version number of the receiver software, yes The valid value is 0-65535 | 2 | 26 |

Note:In the current version,the header length is 28 bytes and the information length is variable; GPS weeks milliseconds for data sampling the moment,not the message coding moment.

### 1.1.2 CRC32 check

The CRC verification can refer to the following example code.

```
const unsigned long ulCrcTable[256] =
{
  0x00000000L, 0x77073096L, 0xee0e612cL, 0x990951baL, 0x076dc419L, 0x706af48fL,
  0xe963a535L, 0x9e6495a3L, 0x0edb8832L, 0x79dcb8a4L, 0xe0d5e91eL, 0x97d2d988L,
  0x09b64c2bL, 0x7eb17cbdL, 0xe7b82d07L, 0x90bf1d91L, 0x1db71064L, 0x6ab020f2L,
  0xf3b97148L, 0x84be41deL, 0x1adad47dL, 0x6ddde4ebL, 0xf4d4b551L, 0x83d385c7L,
  0x136c9856L, 0x646ba8c0L, 0xfd62f97aL, 0x8a65c9ecL, 0x14015c4fL, 0x63066cd9L,
  0xfa0f3d63L, 0x8d080df5L, 0x3b6e20c8L, 0x4c69105eL, 0xd56041e4L, 0xa2677172L,
  0x3c03e4d1L, 0x4b04d447L, 0xd20d85fdL, 0xa50ab56bL, 0x35b5a8faL, 0x42b2986cL,
  0xdbbbc9d6L, 0xacbcf940L, 0x32d86ce3L, 0x45df5c75L, 0xdcd60dcfL, 0xabd13d59L,
  0x26d930acL, 0x51de003aL, 0xc8d75180L, 0xbfd06116L, 0x21b4f4b5L, 0x56b3c423L,
  0xcfba9599L, 0xb8bda50fL, 0x2802b89eL, 0x5f058808L, 0xc60cd9b2L, 0xb10be924L,
  0x2f6f7c87L, 0x58684c11L, 0xc1611dabL, 0xb6662d3dL, 0x76dc4190L, 0x01db7106L,
  0x98d220bcL, 0xefd5102aL, 0x71b18589L, 0x06b6b51fL, 0x9fbfe4a5L, 0xe8b8d433L,
  0x7807c9a2L, 0x0f00f934L, 0x9609a88eL, 0xe10e9818L, 0x7f6a0dbbL, 0x086d3d2dL,
  0x91646c97L, 0xe6635c01L, 0x6b6b51f4L, 0x1c6c6162L, 0x856530d8L, 0xf262004eL,
  0x6c0695edL, 0x1b01a57bL, 0x8208f4c1L, 0xf50fc457L, 0x65b0d9c6L, 0x12b7e950L,
  0x8bbeb8eaL, 0xfcb9887cL, 0x62dd1ddfL, 0x15da2d49L, 0x8cd37cf3L, 0xfbd44c65L,
  0x4db26158L, 0x3ab551ceL, 0xa3bc0074L, 0xd4bb30e2L, 0x4adfa541L, 0x3dd895d7L,
  0xa4d1c46dL, 0xd3d6f4fbL, 0x4369e96aL, 0x346ed9fcL, 0xad678846L, 0xda60b8d0L,
  0x44042d73L, 0x33031de5L, 0xaa0a4c5fL, 0xdd0d7cc9L, 0x5005713cL, 0x270241aaL,
  0xbe0b1010L, 0xc90c2086L, 0x5768b525L, 0x206f85b3L, 0xb966d409L, 0xce61e49fL,
  0x5edef90eL, 0x29d9c998L, 0xb0d09822L, 0xc7d7a8b4L, 0x59b33d17L, 0x2eb40d81L,
  0xb7bd5c3bL, 0xc0ba6cadL, 0xedb88320L, 0x9abfb3b6L, 0x03b6e20cL, 0x74b1d29aL,
  0xead54739L, 0x9dd277afL, 0x04db2615L, 0x73dc1683L, 0xe3630b12L, 0x94643b84L,
  0x0d6d6a3eL, 0x7a6a5aa8L, 0xe40ecf0bL, 0x9309ff9dL, 0x0a00ae27L, 0x7d079eb1L,
  0xf00f9344L, 0x8708a3d2L, 0x1e01f268L, 0x6906c2feL, 0xf762575dL, 0x806567cbL,
  0x196c3671L, 0x6e6b06e7L, 0xfed41b76L, 0x89d32be0L, 0x10da7a5aL, 0x67dd4accL,
  0xf9b9df6fL, 0x8ebeeff9L, 0x17b7be43L, 0x60b08ed5L, 0xd6d6a3e8L, 0xa1d1937eL,
  0x38d8c2c4L, 0x4fdff252L, 0xd1bb67f1L, 0xa6bc5767L, 0x3fb506ddL, 0x48b2364bL,
  0xd80d2bdaL, 0xaf0a1b4cL, 0x36034af6L, 0x41047a60L, 0xdf60efc3L, 0xa867df55L,
  0x316e8eefL, 0x4669be79L, 0xcb61b38cL, 0xbc66831aL, 0x256fd2a0L, 0x5268e236L,
```

```
  0xcc0c7795L, 0xbb0b4703L, 0x220216b9L, 0x5505262fL, 0xc5ba3bbeL, 0xb2bd0b28L,
  0x2bb45a92L, 0x5cb36a04L, 0xc2d7ffa7L, 0xb5d0cf31L, 0x2cd99e8bL, 0x5bdeae1dL,
  0x9b64c2b0L, 0xec63f226L, 0x756aa39cL, 0x026d930aL, 0x9c0906a9L, 0xeb0e363fL,
  0x72076785L, 0x05005713L, 0x95bf4a82L, 0xe2b87a14L, 0x7bb12baeL, 0x0cb61b38L,
  0x92d28e9bL, 0xe5d5be0dL, 0x7cdcefb7L, 0x0bdbdf21L, 0x86d3d2d4L, 0xf1d4e242L,
  0x68ddb3f8L, 0x1fda836eL, 0x81be16cdL, 0xf6b9265bL, 0x6fb077e1L, 0x18b74777L,
  0x88085ae6L, 0xff0f6a70L, 0x66063bcaL, 0x11010b5cL, 0x8f659effL, 0xf862ae69L,
  0x616bffd3L, 0x166ccf45L, 0xa00ae278L, 0xd70dd2eeL, 0x4e048354L, 0x3903b3c2L,
  0xa7672661L, 0xd06016f7L, 0x4969474dL, 0x3e6e77dbL, 0xaed16a4aL, 0xd9d65adcL,
  0x40df0b66L, 0x37d83bf0L, 0xa9bcae53L, 0xdebb9ec5L, 0x47b2cf7fL, 0x30b5ffe9L,
  0xbdbdf21cL, 0xcabac28aL, 0x53b39330L, 0x24b4a3a6L, 0xbad03605L, 0xcdd70693L,
  0x54de5729L, 0x23d967bfL, 0xb3667a2eL, 0xc4614ab8L, 0x5d681b02L, 0x2a6f2b94L,
  0xb40bbe37L, 0xc30c8ea1L, 0x5a05df1bL, 0x2d02ef8dL
};


unsigned int GetCrc32(const unsigned char* pBuffer, int iLen)
{
    unsigned int crc = 0;

    for (int n = 0; n < iLen; n++)
    {
       Crc = ulCrcTable[(crc ^ pBuffer[n]) & 0xff] ^ (crc >> 8);
    }

    return crc;
}
```

### 1.1.3 satellite navigation raw data message

Message ID: 2365

Get instruction: insrtk gnssrawdata [param]

param configurable parameters 0-4,0 is no output ,1-4 output COM port.

For example,insrtk gnssrawdata 2 outputs this message data on com2 port.

| field | content | type | description | byte | offset |
|---|---|---|---|---|---|
| 1 | Head | | Message header | H | 0 |
| 2 | FiqCount | uint | FIQ count | 4 | H |
| 3 | Quality | uint | GNSS localization status | 4 | H+4 |

| | | | | | |
|---|---|---|---|---|---|
| 4 | DoPvtRtk | uchar | RTK&PVT the solution title | 1 | H+8 |
| 5 | nSm | uchar | The number of satellite | 1 | H+9 |
| 6 | HDOP | double | Flat position geometric Precision factor | 8 | H+10 |
| 7 | time_tag | double | GPS within seconds (s) | 8 | H+18 |
| 8 | Lat | double | Latitude (deg) | 8 | H+26 |
| 9 | Lon | double | Longitude (deg) | 8 | H+34 |
| 10 | Hgt | double | Ellipsoidal height (m) | 8 | H+42 |
| 11 | VelNorth | double | Northbound speed (m/s) | 8 | H+50 |
| 12 | VelEast | double | East direction velocity (m/s) | 8 | H+58 |
| 13 | VelDown | double | Ground velocity (m/s) | 8 | H+66 |
| 14 | LatVar | double | Latitudinal position variance (m2) | 8 | H+74 |
| 15 | LonVar | double | Longitude position variance (m2) | 8 | H+82 |
| 16 | HgtVar | double | Elevation position variance (m2) | 8 | H+90 |
| 17 | VelNorthVar | double | Northbound velocity variance (m2/s2) | 8 | H+98 |
| 18 | VelEastVar | double | Eastbound velocity variance (m2/s2) | 8 | H+106 |
| 19 | VelDownVar | double | Ground velocity variance (m2/s2) | 8 | H+114 |
| 20 | CRC | uint | CRC32 check | 4 | H+122 |

### 1.1.4 Raw data message of the inertial sensor

**Message ID**: 2364

Get instruction: insrtk imurawdata [param]

Param configurable parameters 0-4,0 is no output ,1-4 output COM port.

For example,insrtk imurawdata 2 outputs this message data on com2 port.

| field | content | type | description | byte | offset |
|---|---|---|---|---|---|
| 1 | Head | | Message header | H | 0 |
| 2 | imu_id | uchar | The IMU identification bit | 1 | H |

| | | | | | |
|---|---|---|---|---|---|
| 3 | FiqCount | uint | FIQ count | 4 | H+1 |
| 4 | dt | double | Sampling interval(s) | 8 | H+5 |
| 5 | time_tag | double | GPS seconds within the week (s) | 8 | H+13 |
| 6 | rawIMUData | uchar[15] | The IMU raw data | 15 | H+21 |
| 7 | CRC | uint | CRC32 Check | 4 | H+36 |

### 1.1.5 Tilt measurement solution data message

The K803 module uses satellite navigation raw data and inertial sensor raw data to perform tilt measurement calculations.The output solution data mainly includes: positioning solution status, antenna phase center latitude and longitude height, ENU speed, Euler Angle, tilt angle, ENU position compensation data calculated from the antenna phase center to the centering pole tip, position compensation Data validity indication parameter, tilt measurement status indication parameter, initialization progress indication parameter, initialization Shake/rotation indication parameters, shake indication parameters required for attitude divergence, and error codes. Users can use the antenna phase center latitude and longitude height and ENU position reduction compensation data to achieve the tilt measurement function.

**Message ID**: 2366

Get instruction: log [param1] insrtk compensate ontime [param2]

param 1 com port, optional configuration, default com1; param2,message output interval, support maximum 20 Hz output.

For example, log com2 insrtkcompensate ontime 0.2, to output this message data at 0.2-second intervals on the com2 port.

| field | content | type | description | byte | offset |
|---|---|---|---|---|---|
| 1 | Head | | Message header | H | 0 |
| 2 | PosMode | uchar | Locate the solution state | 1 | H |
| 3 | Lat | double | Latitude (deg) | 8 | H+1 |
| 4 | Lon | double | Longitude (deg) | 8 | H+9 |
| 5 | Hgt | double | altitude (m) | 8 | H+17 |
| 6 | VelEast | float | East-direction velocity (m / s) | 4 | H+25 |
| 7 | VelNorth | float | Northbound speed (m / s) | 4 | H+29 |

| | | | | | |
|---|---|---|---|---|---|
| 8 | VelUp | float | Skyward velocity (m / s) | 4 | H+33 |
| 9 | Pitch | float | Pitch angle (deg) | 4 | H+37 |
| 10 | Roll | float | Roll Angle(deg) | 4 | H+41 |
| 11 | Heading | float | heading angle(deg) | 4 | H+45 |
| 12 | Tilt Angle | float | Tilting angle (deg) | 4 | H+49 |
| 13 | Compensate Valid | uchar | Is the position compensation parameter valid, Effect, 0: invalid, 1: valid | 1 | H+53 |
| 14 | DeltaEast | double | East-facing position compensation (m) | 8 | H+54 |
| 15 | DeltaNorth | double | Northbound position compensation (m) | 8 | H+62 |
| 16 | DeltaUp | double | Sky position compensation (m) | 8 | H+70 |
| 17 | InsRtkStatus | char | Ttilt measurement state indication | 1 | H+78 |
| 18 | StepProcess | ushort | Initialize the progress parameter (01000, unit:1/1000) | 4 | H+79 |
| 19 | DirectionStatus | uchar | Shake and rotate switching indication during installation angle calibration | 1 | H+83 |
| 20 | NeedAShack | uchar | Indication of whether the filter is divergent: 0: filter re-converges, 1: filter divergence | 1 | H+84 |
| 21 | ErrorCode | uint | Error code | 4 | H+85 |
| 22 | Reserved | | reserved | 4 | H+89 |
| 23 | CRC | uint | CRC32 Check | 4 | H+93 |

Note: Whether the Compensate Valid field is valid is related to the Error Code field. In working mode, except for the RTK fixed solution unstable error state, if other error states exist, the field indication value is invalid. Error Code Field description is detailed in Section 1. 1.9.

### 1.1.6 Description of tilt measurement status

InsRtkStatus The parameter indicates the working stage of the current tilt measurement, including waiting for initialization,static data processing, dynamic convergence, and convergence completion into the working mode.

| InsRtkStatus | | |
|---|---|---|
| name | Numeric value | description |
| WAIT_INIT | -1 | Wait to receive the initialization instruction status |
| MOTIONLESS | 1 | Received initialization command, static data processing is in progress, keep the receiver stationary |
| SHACK | 2 | Enter the dynamic convergence phase, shaking back and forth Receive machine |
| COLIBRATE | 3 | If it is the installation angle error calibration mode,Shake or rotate the receiver according to the Direction Status instructions,If it is in FAST_INIT mode, the Direction Status always prompts to shake the receiver, and there will be no rotation status indication. |
| INITOK | 4 | Enter the working mode |

### 1.1.7 Instructions for shaking and rotating switching during installation angle error calibration process

The Direction Status parameter indication is only valid when InsRtkStatus is in the COLIBRATE phase.

| DirectionStatus | | |
|---|---|---|
| name | Numeric value | description |
| SHACK | 1 | Shake the receiver back and forth |
| ROT | 2 | Slowly rotate the receiver until it cuts to the SHACK state |

### 1.1.8 Positioning solution status description

| PosMode | | |
|---|---|---|
| name | Numeric value | description |
| PVT | 1 | Single point positioning |
| RTD/SBAS | 2 | Pseudo range difference or SBAS positioning |
| RTK FIXED | 4 | RTK fixed solution |

| | | |
|---|---|---|
| RTK FLOAT | 5 | The RTK floating- point solution |
| DR | 6 | Combination navigation solution |

### 1.1.9 Error code description

| ErrorCode | | |
|---|---|---|
| name | numeric value | description |
| ERROR_GYRO_OUTRANGE | 0x0001 | The gyroscope data excess program sends a reset command (see Section 1.2 for details) to reset the tilt measurement module and then sends an initialization command to re-initialize. |
| ERROR_ACC_OUTRANGE | 0x0002 | It only prompts when the accelerometer data exceeds the range and no other operations are required by the user. |
| ERROR_RTK_UNSTABLE | 0x0004 | RTK fixed state is unstable. If the module automatically resets in the MOTIONLESS stage, an initialization command needs to be sent. If it only prompts in other stages, no other operations are required by the user. |
| ERROR_INITIALFAILED | 0x0010 | Initialization failed The module is automatically reset and needs to send the initialization instruction Re initialization. |
| ERROR_CENTERINGROD_LENGTH | 0x0100 | If the deviation between the centering rod length estimate and the input is too large, you need to check the centering rod input and send a reset command (see section 1.2 for details). After resetting the tilt measurement module, send an initialization command to re-estimate. |
| ERROR_MISALIGNMENTANGLE | 0x0200 | If the IMU module installation angle estimation error is too large, a reset command must be sent (see Section 1.2 for details).After resetting the tilt measurement module, send an initialization command to re-initialize. |
| ERROR_ATTITUDE | 0x1000 | The filter diverges |

| | | |
|---|---|---|
| | | The user is required to shake the receiver until NeedAShack indicates that the filter has re converged. |
| ERROR_RTK_FIXEDERR | 0x2000 | RTK fixed solution is wrong |
| ERROR_NOTSTATIC | 0x4000 | The receiver is not stationary only in the MOTIONLESS stage. It only needs to prompt the user to keep the receiver stationary and no other operations are required. |

Note: The error codes listed in the above table may exist in ErrorCode at the same time.

### 1.1.10 Tilt measurement compensation data usage

Tilt measurement compensation data using sample code.

```
/*Input parameters:
PosAnt: Solve the coordinates of the antenna phase center longitude and latitude
DeltaENU: Solve the position compensation (DeltaEast, DeltaNorth, DeltaUp) data in the data message
output parameters:
PosRodTip: The coordinates of the longitude and latitude of the centering rod tip (Lat, Lon, Hgt)
*/

void GetRodTipPosition(const double PosAnt[3], const double DeltaENU[3],
                       double PosRodTip[3])
{
    #define PI (3.1415926535897932)
    const double f = 1 / 298.257223563;
    const double Re = 6378137.0;
    double e2 = 2.0 * f - f * f;
    double W = 0.0;
    double Rm = 0.0, Rn = 0.0;
    double lon_rad = 0.0, lat_rad = 0.0;
    double sinlat = 0.0, sinlat2 = 0.0;

    lat_rad = PosAnt[0] * PI / 180.0;
    lon_rad = PosAnt[1] * PI / 180.0;
    sinlat = sin(lat_rad);
    sinlat2 = sinlat * sinlat;
    W = sqrt(1 - e2 * sinlat2);
    Rm = Re * (1 - e2) / (W * W * W);
    Rn = Re / W;

    PosRodTip[0] = lat_rad + DeltaENU[1] / (Rm + PosAnt[2]);
```

```
    PosRodTip[1] = lon_rad + DeltaENU[0] / (Rn + PosAnt[2]) / cos(lat_rad);
    PosRodTip[2] = PosAnt[2] + DeltaENU[2];
    PosRodTip[0] = PosRodTip[0] / PI * 180.0;
    PosRodTip[1] = PosRodTip[1] / PI * 180.0;
}
```

## 1.2 configuration instruction set

### 1.2.1 GNSS configuration instruction set

Refer to the K803 Board Card User Manual for this instruction set

### 1.2.2 Inclination measurement configuration instruction set

| Directive overview | |
|---|---|
| insrtk reset | The tilt measurement module is reset |
| insrtk saveconfig | Save the configuration parameters |
| insrtk showconfig | Obtain the tilt measurement module configuration information |
| insrtk freset | Clear the user configuration parameters |
| tst ins debug [param] | Configure the Debug information output Default 0 does not output; 2 output |
| insrtk rodlength [param] | Configure the centering rod length Unit: mm (0,20000] |
| insrtkleverarmgnss [param1] [param2] [param3] | Configure the GNSS antenna phase center rod arm vector Unit: mm (-500,500) |
| insrtk leverarmarp [param1] [param2] [param3] | Configure the receiver ARP rod arm vector<br>Unit: mm (-500,500) |
| insrtk installangle [param1] [param2] [param3] | Configure the IMU module installation corner Unit: Degrees [- 180, 180] |
| | Configure the RTK solution frequency<br>Default 5 Hz parameter, do not change the configuration |
| insrtk fastinit [on/off] | The configuration fast initialization (FAST_INIT) mode is off by default, that is, the installation angle error calibration mode |

| | |
|---|---|
| insrtk spattitudeth [param1] [param2] | Configure the threshold of heading angle (param1) and attitude angle (param2) for filter convergence in fast initialization mode.<br><br>Unit: Degree (0, 5] Default 1.5, 1.5Parameter |
| insrtk polexy on [param1] [param2] | In manual configuration quick initialization mode, the lever arm vector plane coordinates corrected by the IMU module installation angle error<br><br>Unit: meters (-0.1, 0.1) |
| insrtk shakeparam [param1] [param2] [param3] | Configure the IMU module installation angle error calibration mode, the number of shaking groups (param1), the number of rotation directions (param2), and the number of shaking per group (param3)<br><br>param1: (0, 10], param2: (0, 10]<br><br>param3: (0, 30]<br><br>Default 2, 2, 5 parameters |
| insrtk init 1 | The tilt measurement module initialization instruction |

## 1.3 Detailed description of the tilt measurement configuration instruction

### 1.3.1 insrtk showconfig Instruction

Obtain the tilt measurement module configuration information, with the following contents.

1. INSRTK Version

Version number information, which contains 18 uchar bytes.

M. S. sss, ee, yymmdd

M: Main Revision [0 ~ 9]

S: Stage Infor 0: Development Stage

1: Alpha Release Stage

2: Beta Release Stage

3: Production Release Stage

4: QFE Version

sss: Sub Revision [000 ~ 999]

ee: Engineer Revision

yymmdd: year, month, day

2. Install Angle

MU module installation angle.

3. GNSS Lever Arm

GNSS antenna phase center rod arm vector.

4. ARP Lever Arm

The ARP rod-arm vector.

5. Centering Rod

for middle rod length

6. InitFast

Quick initialization switch.

7. SPAttTH

In fast initialization mode, the heading angle and the attitude angle threshold converge.

8. Pole XY

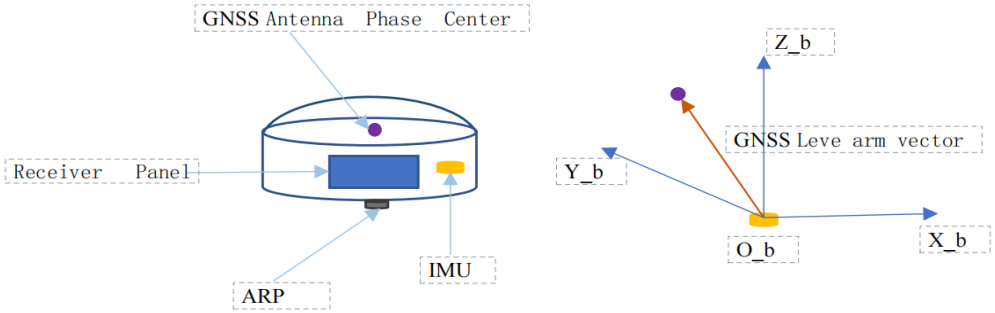
Arm vector plane coordinates after mounting angle error compensation in fast initialization mode.

9. Shake Rotation

Calibration installation angle mode, number of shaking groups, number of rotation directions, and number of shaking of each group.

### 1.3.2 insrtk leverarmgnss instruction

Configure the GNSS antenna phase center arm vector, the unit is millimeters, the value range is (-500, 500).



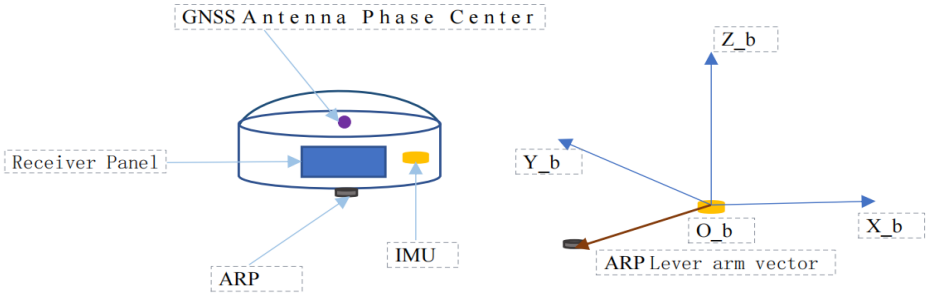

For example, insrtk leverarmgnss -10 10 50, configure the GNSS antenna phase center lever arm vector coordinates to (-10, 10, 50) mm.

Before using the tilt measurement module, the user needs to define the carrier coordinate system (b system) based on the actual receiver design. Its origin is located at the center of the IMU (the corner position of the K803 module shield with perforations), which is the right-hand coordinate system O-XYZ. As shown in the figure above, a typical receiver design example, the carrier coordinate system is defined as follows: facing the receiver panel, the Y-axis is perpendicular to the panel and outward is positive, the Z-axis is perpendicular to the receiver plane and upward is positive, and the X-axis is perpendicular to The O-YZ plane forms the right-handed coordinate system.

The tilt measurement data is actually obtained by fusion and calculation of the GNSS and IMU sensor data. Normally, the GNSS antenna phase center position does not coincide with the IMU center position. When the receiver rotates, the position and velocity measured by the GNSS There is a difference from what the IMU senses, this phenomenon is called the lever arm effect. In order to eliminate the influence of the lever arm effect, the user is required to configure the GNSS antenna phase center lever arm vector. The GNSS antenna phase center rod arm vector is defined as: pointing from the IMU center to the GNSS antenna phase The projection of the vector formed by the center in the carrier coordinate system.

### 1.3.3 insrtk leverarmarp instruction

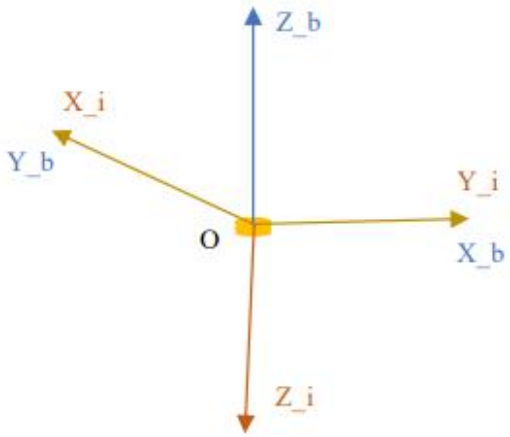
Configure the ARP lever arm vector, the unit is millimeters, the value range is (-500, 500).
For example, insrtk leverarmarp - 10 10 -30 configures the ARP lever arm vector coordinates as (- 10, 10, -30) mm




The ARP arm vector is defined as: the projection of the vector formed by the center of the IMU (the corner position of the K803 module shield with holes) pointing to the ARP point of the receiver in the carrier coordinate system.

### 1.3.4 insrtk installangle instruction

Configure the IMU module installation angle, the unit is degrees, the value range is [-180, 180].




Before using the tilt measurement module, the user needs to configure the installation angle according to the relative installation relationship between the actual IMU module and the receiver main board. The IMU module installation angle represents the rotation relationship between the carrier coordinate system (b system) and the IMU coordinate system (i system). The IMU coordinate system is defined as: the origin is located at the center of the IMU (the K803 module shield has a perforated corner position), and each axis coincides with its corresponding IMU sensitive axis (the module shield has a sensitive axis indication), forming a right-hand coordinate Tie. The carrier coordinate system rotates around its Z-axis, X-axis, and Y-axis in sequence and then coincides with the IMU coordinate system. Each coordinate axis rotates counterclockwise to be positive, and the rotation angles correspond to installation angle 3, installation angle 1, and installation angle 2 respectively.

In the installation relationship example shown in the figure above, if installation angle 1 is 180 degrees, installation angle 2 is 0 degrees, and installation angle 3 is +90 degrees, the insrtk installangle 180 0 90 command should be sent for installation angle configuration.

### 1.3.5 insrtk fastinit instruction

Configure initialization mode, off: IMU installation angle error calibration mode, on: quick initialization mode, default off is installation angle error calibration mode.

For example, insrtk fastinit on is configured in fast initialization mode.

When fast initialization mode is turned on, the initialization process only requires the user to shake the receiver。

If you need to calibrate the installation angle error of the IMU module, you need to turn off the quick initialization mode (configured as the installation angle error calibration mode). After the calibration is successful, the lever arm vector plane coordinates (PoleXY) after the installation angle error compensation will be automatically stored in FLASH. , PoleXY can be automatically obtained every time the device is turned on to prepare for quick initialization. The user can also send the insrtk polexy on command to manually configure the compensation value, but it is generally not recommended to use this command to configure the compensation value manually. In addition, if the receiver is used for the first time, or the measurement accuracy of the receiver is significantly different from the nominal accuracy, it is recommended that the user calibrate the error before use.

## 1.4 Inclination measurement configuration process

1. insrtk installangle [param1] [param2] [param3]; //Configuring the IMU module mounting angle

2. insrtkleverarmgnss [param1] [param2] [param3]; //Configuring the GNSS Antenna Phase Center Rod Arm

3. insrtk leverarmarp [param1] [param2] [param3]; //Configure the receiver ARP arm vector /*The above configuration information is the basic hardware parameter configuration information. After configuration, send insrtk saveconfig to save and restart the receiver. If the relevant hardware parameters of the receiver have not changed, it does not need to be configured every time*/

/* After items 1, 2, and 3 have been configured and saved, and the receiver has been restarted, perform the following configuration*/

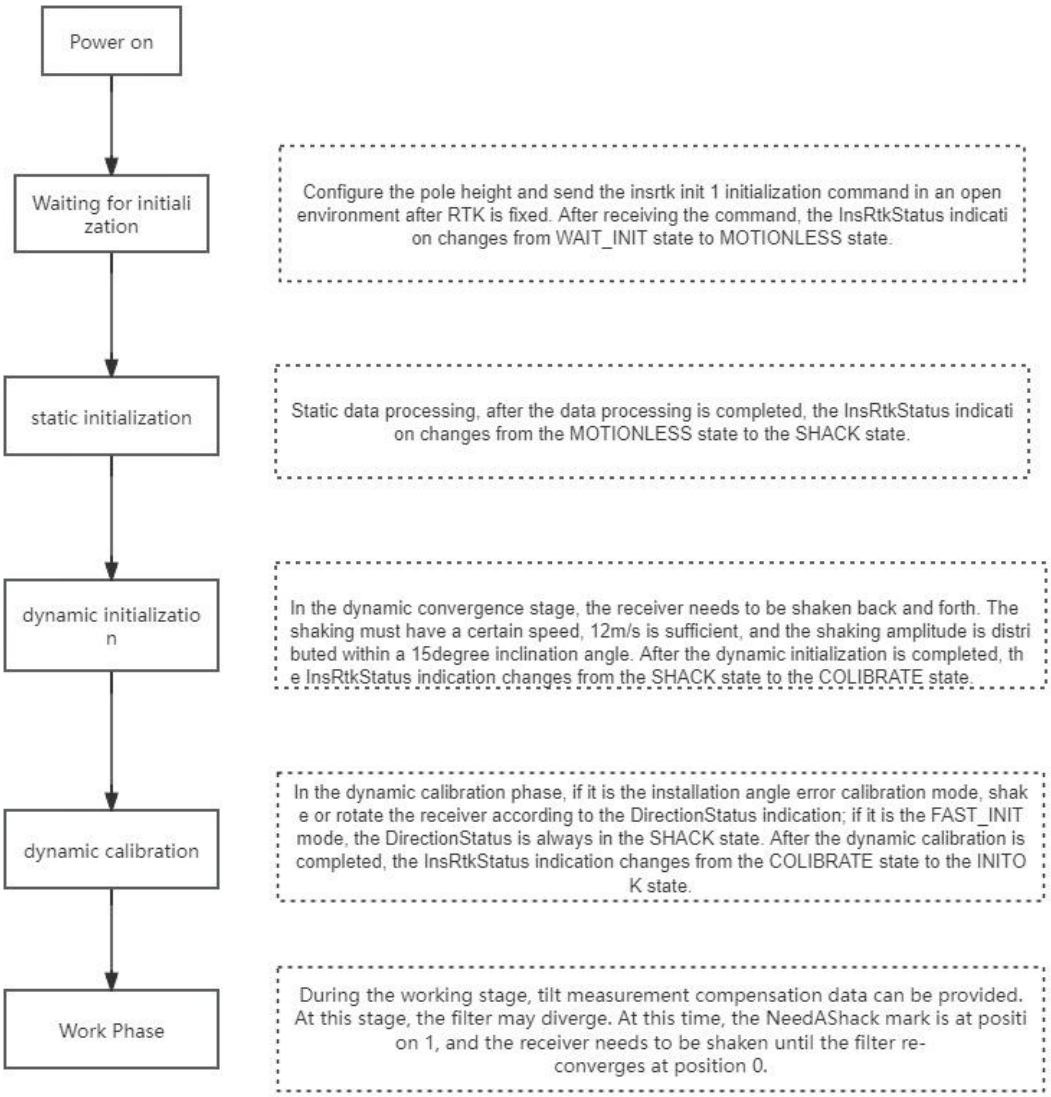
4. insrtk fastinit [on/off]; //Configure the initialization mode. Just complete the configuration before the initialization starts. It will take effect in real time and support configuration saving.

5. insrtk spattitudeth [param1] [param2]; //This configuration only works in fast initialization mode. You only need to complete the configuration before the initialization starts. It will take effect in real time. Configuration saving is supported.

6. insrtk shakeparam [param1] [param2] [param3]; //This configuration only applies to the IMU module installation angle calibration mode. You can complete the configuration before the initialization starts. Configuration saving is supported.

7. insrtk init 1; //Start initialization and enter the working phase after initialization is completed.

## 1.5 Instructions for the actual use process of tilt measurement




Special note: Each stage needs to obtain and immediately process the error code indication of the response ErrorCode. If the inclination measurement module needs to be reset, send the insrtk reset reset command and wait for 1 s before sending the insrtk init 1 initialization command again.
Perform initialization steps.